

Midterm Exam 1 · Fall 2021

MATH 411 · October 1, 2021

Retypeset past exam for practice. The original was closed book and worth 25 points.

1. Let $a, b, c, d \in \mathbb{R}$ satisfy $ad - bc = 1$, and set

$$f(z) = \frac{az + b}{cz + d}, \quad \mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}.$$

- (a) **(2 points)** Show that f maps $\mathbb{H}$ into $\mathbb{H}$.
- (b) **(2 points)** Is $f : \mathbb{H} \rightarrow \mathbb{H}$ holomorphic?
- (c) **(5 points)** A line or circle can be given by an equation

$$Ax + By + C(x^2 + y^2) = D,$$

where $A, B, C, D \in \mathbb{R}$ and $z = x + iy$. For the choice $(a, b, c, d) = (0, -1, 1, 0)$, show that $f(z) = -1/z$ sends the portion of such a line or circle in $\mathbb{H}$ to the portion of a line or circle in $\mathbb{H}$.

2. Consider the real function

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

- (a) **(3 points)** Show that f is infinitely differentiable at 0 and compute every $f^{(k)}(0)$. You may use that a rational function of x multiplied by e^{-1/x^2} tends to 0 as $x \rightarrow 0$.
- (b) **(3 points)** Prove or disprove: an analytic function on a complex neighborhood of 0 agrees with f on the real axis near 0.

3. (10 points) Suppose

$$f(T) = 1 + a_1T + a_2T^2 + \cdots$$

has positive radius of convergence and every a_k is a positive real number. The formal inverse $g(T) = \sum_{n \geq 0} b_n T^n \in \mathbb{C}[[T]]$ satisfies $f(T)g(T) = 1$. Prove that g also has positive radius of convergence.

Hint. First find a recurrence for b_n , then choose $0 < \delta < 1$ such that $\sum_{k \geq 1} a_k \delta^k < 1$.